

# Property rights, Fiscal Capacity, and Social Capacity: The Lasting Impact of the Taiping Rebellion

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  - “*The social and institutional legacies of conflict are arguably the most important but least understood of all war impacts.*” (Blattman and Miguel, 2010, “Civil Wars”)
- **This paper:** provides evidence on **institutional and development effects** of the Taiping Rebellion in Qing China (1851-1864) over a long run.
  - The Rebellion facilitated long-term development via **strengthening property rights, local fiscal capacity, and social capacity.**

# Taiping Rebellion (1850-1864)

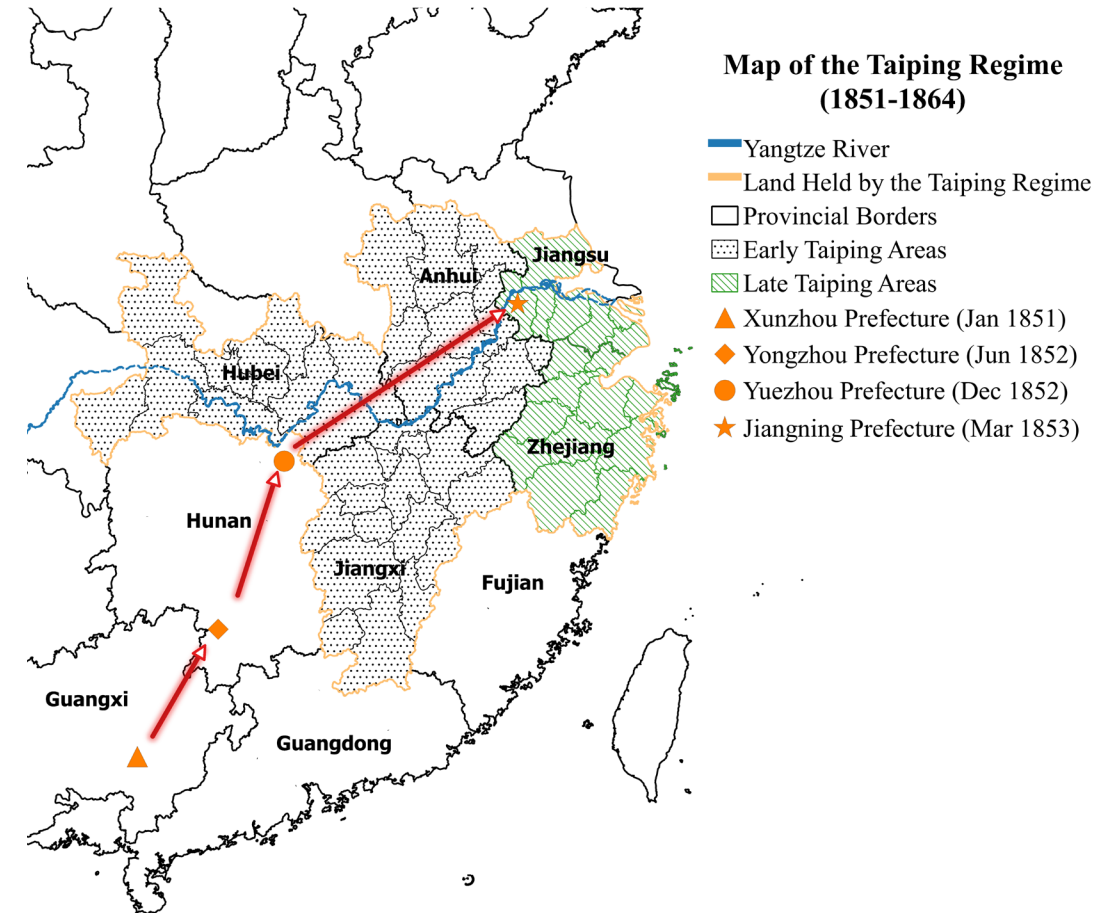
- Deadliest civil war in human history (Ho, 1959)
  - Aside: “Taiping” means “great peace” in Chinese.

|   | War ⇅                          | Death range ⇅                            | Date ⇅    |
|---|--------------------------------|------------------------------------------|-----------|
| 1 | World War II                   | 70–85 million <sup>[2][3][4]</sup>       | 1939–1945 |
| 2 | Taiping Rebellion              | 20–70 million <sup>[5][6][7][8]</sup>    | 1850–1864 |
| 3 | Mongol invasions and conquests | 20–60 million <sup>[9][10][11][12]</sup> | 1206–1368 |

*Wikipedia: List of wars by death toll*

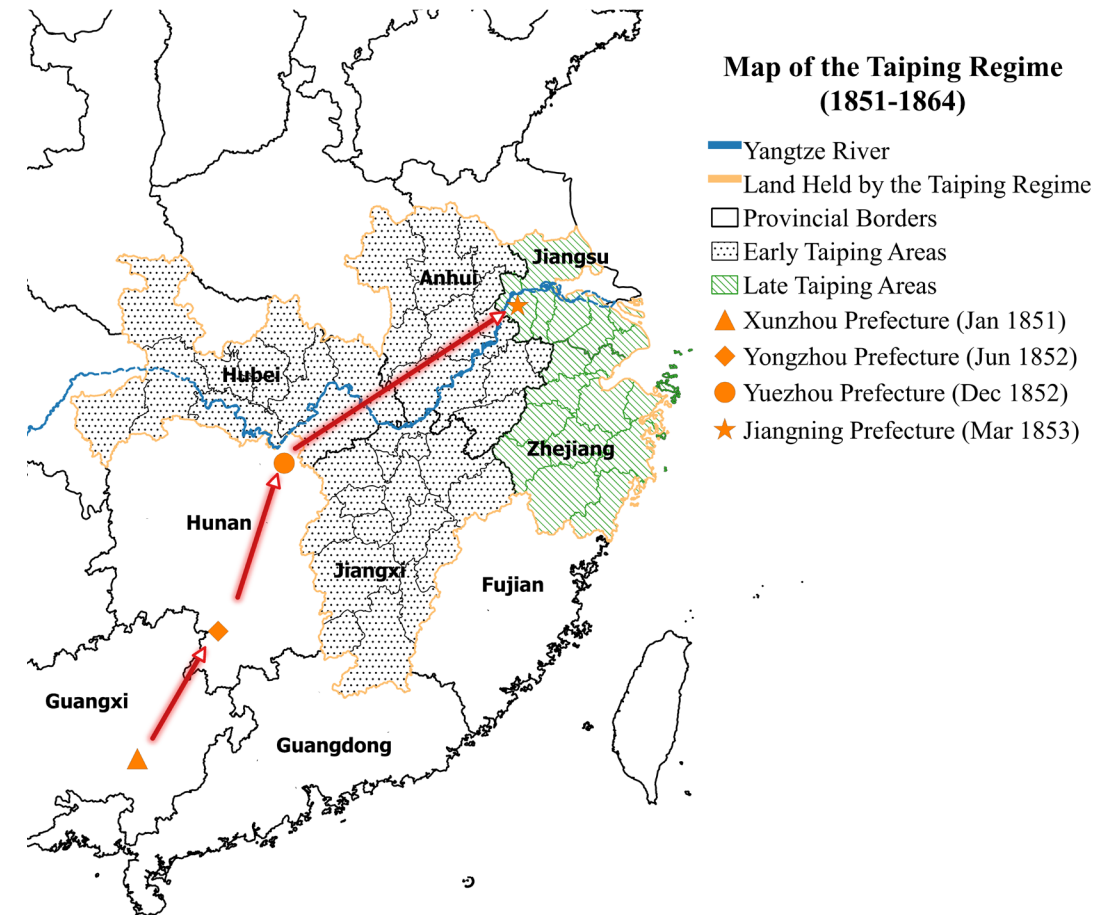
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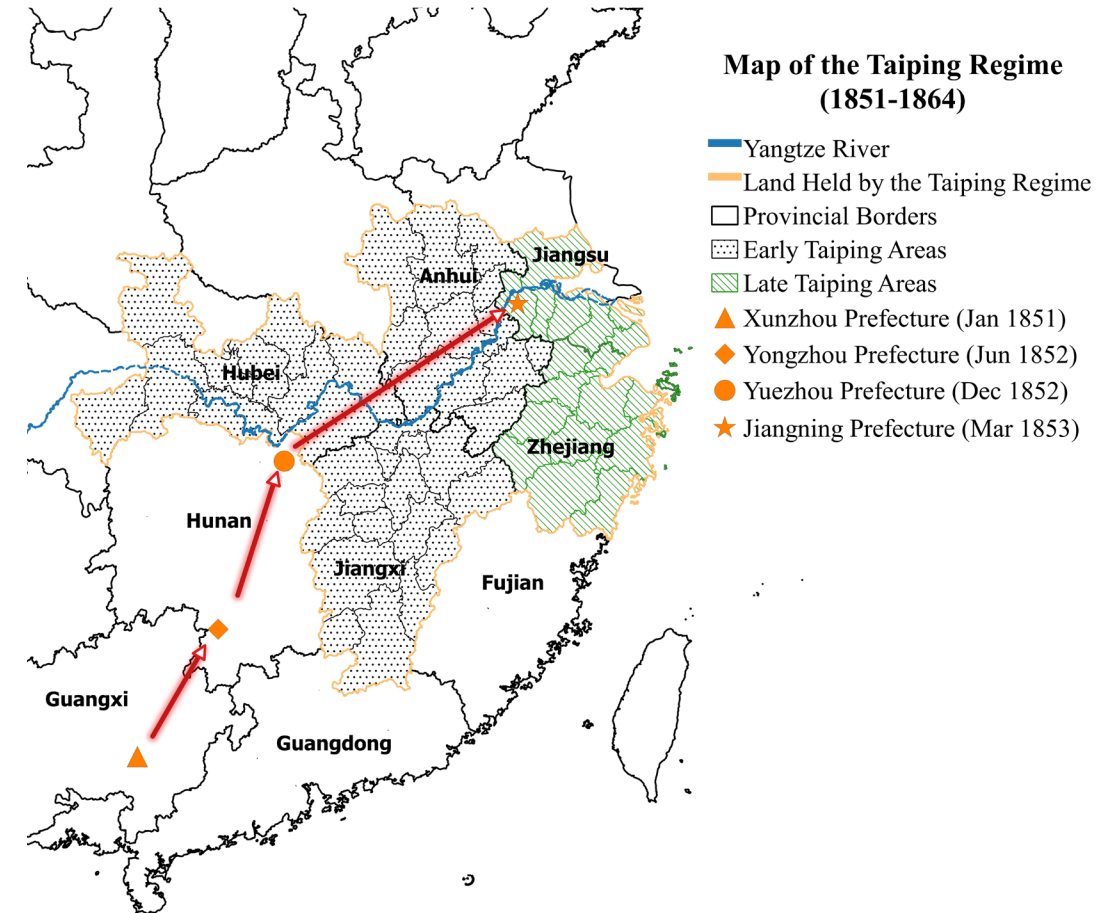
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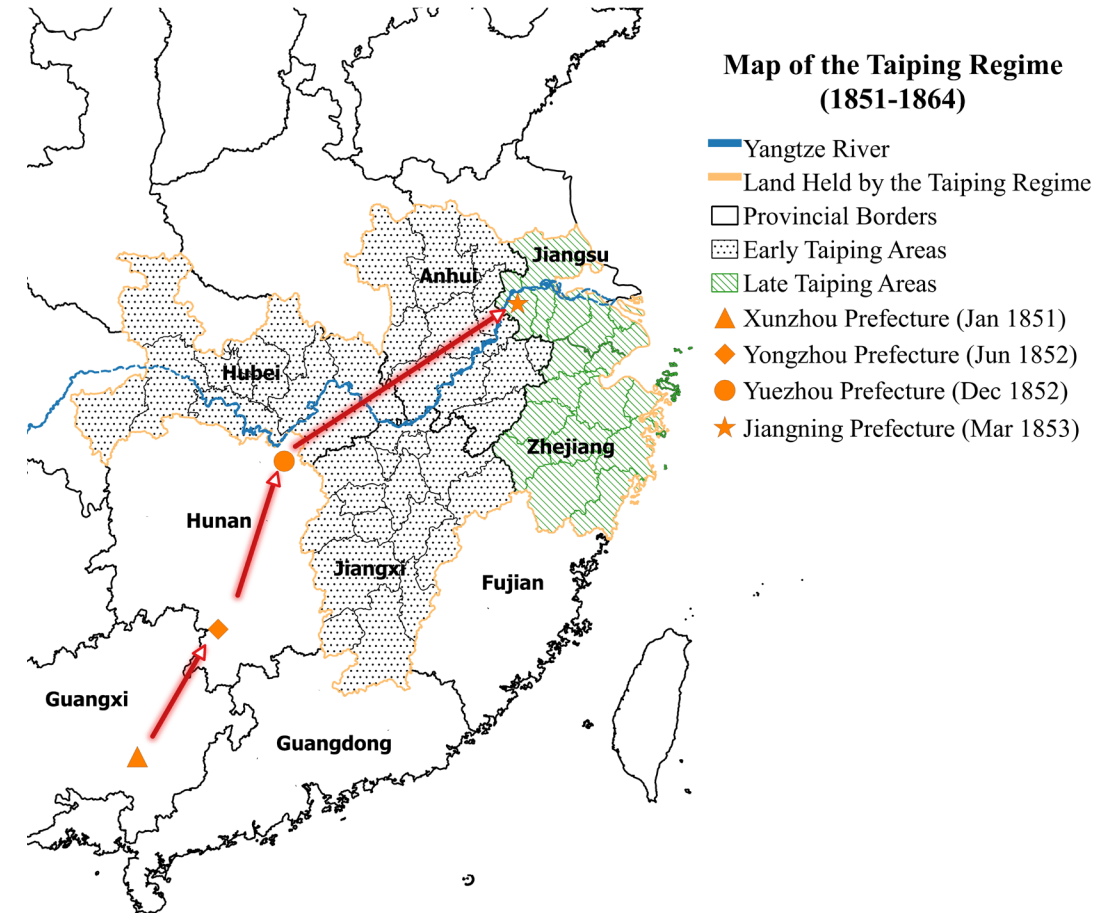
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    - The regime established its capital here (Nanjing today).
    - Much more fierce battles.



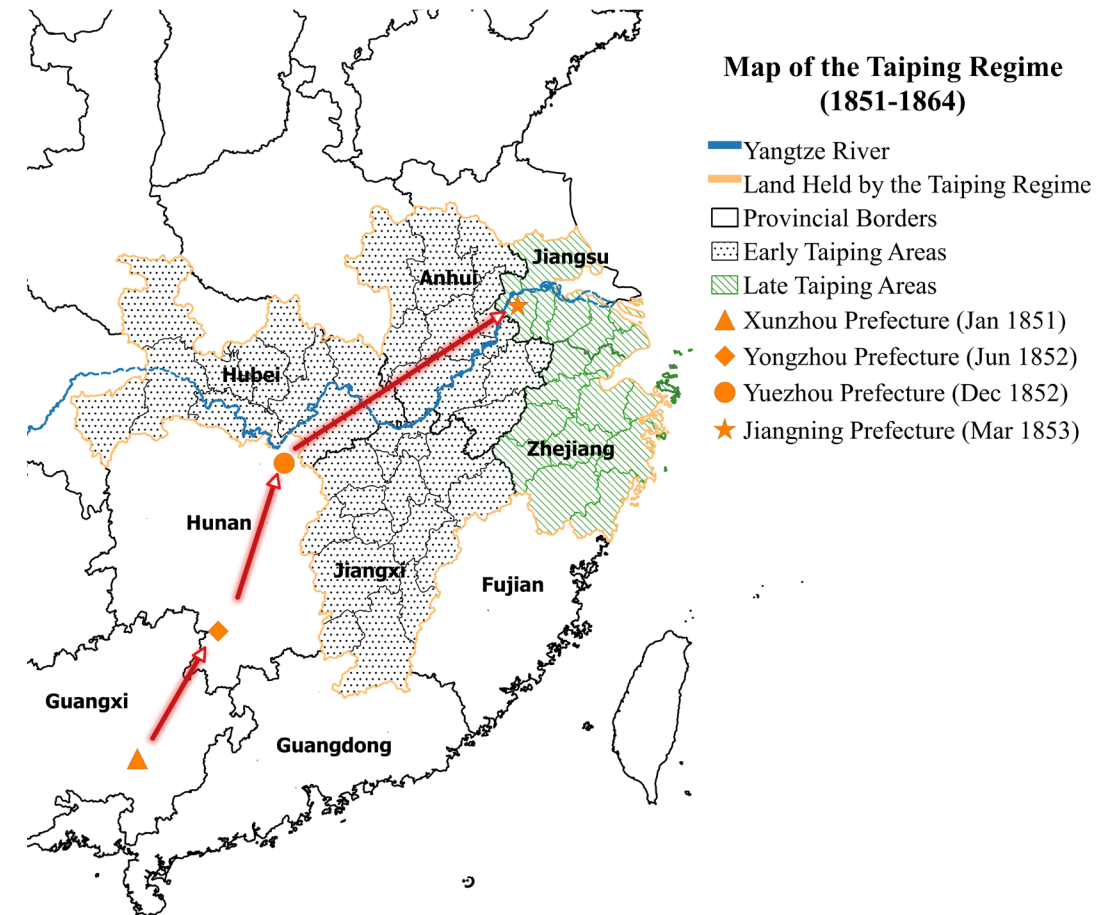
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    - Much more fierce battles.
- **The Taiping regime's territorial control was *incomplete*.**
  - Some areas remained under the Qing forces, and the control frequently changed hands between the two sides.



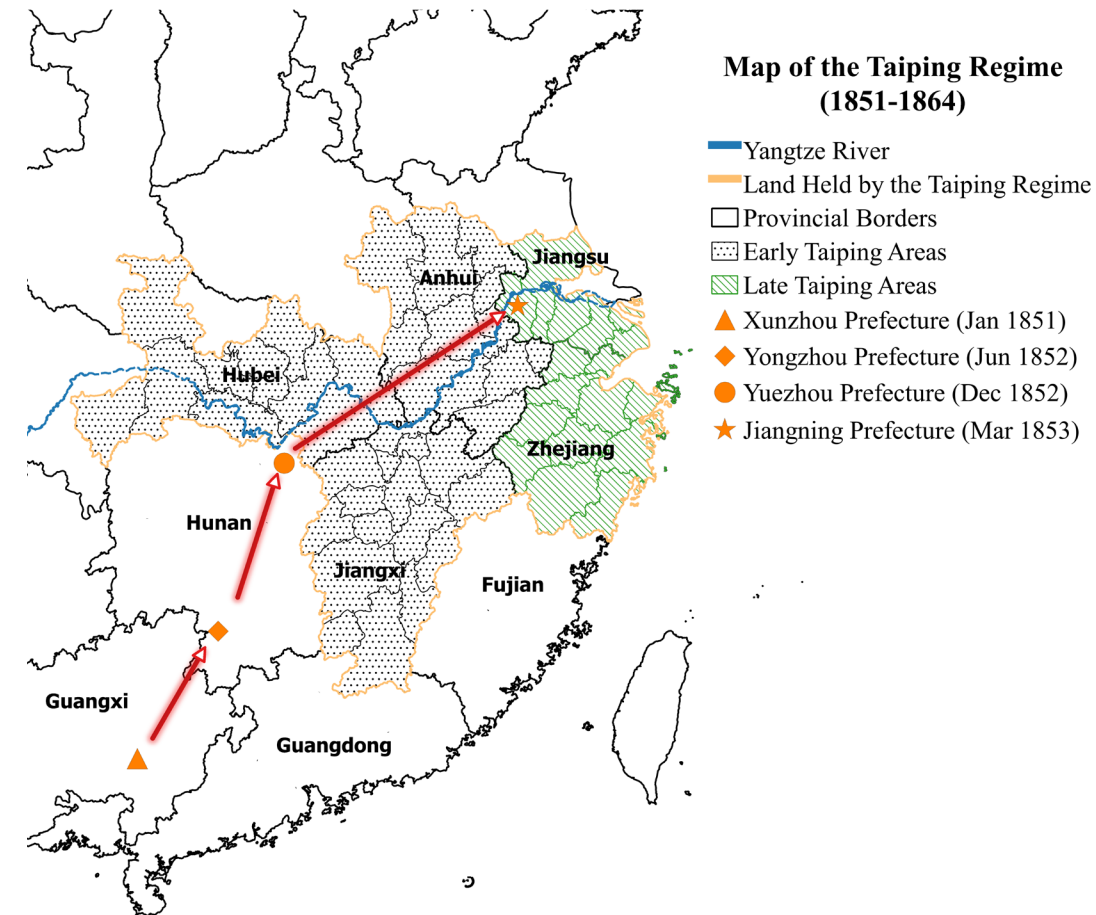
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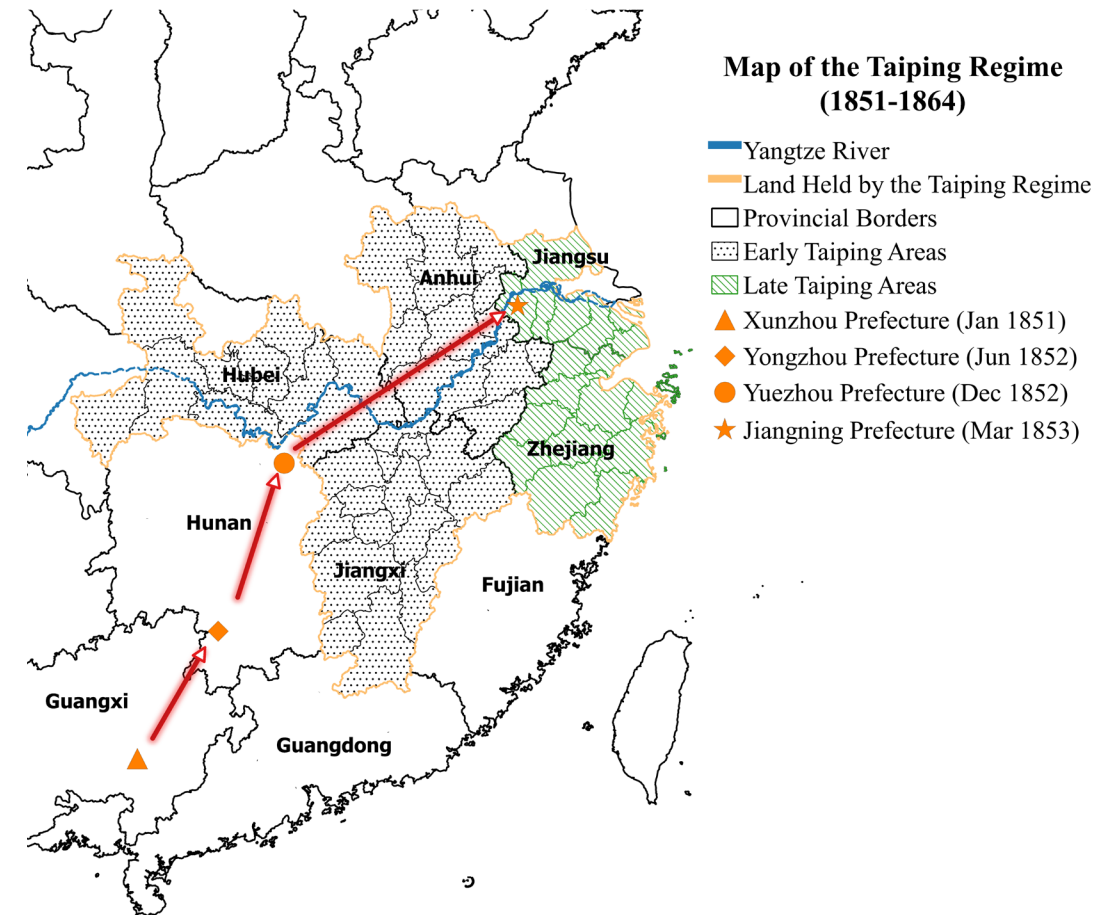
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  - In **Late Taiping areas**, the Taiping regime redistributed land to farmers to finance war, and the redistribution was largely respected after the fall of the regime.



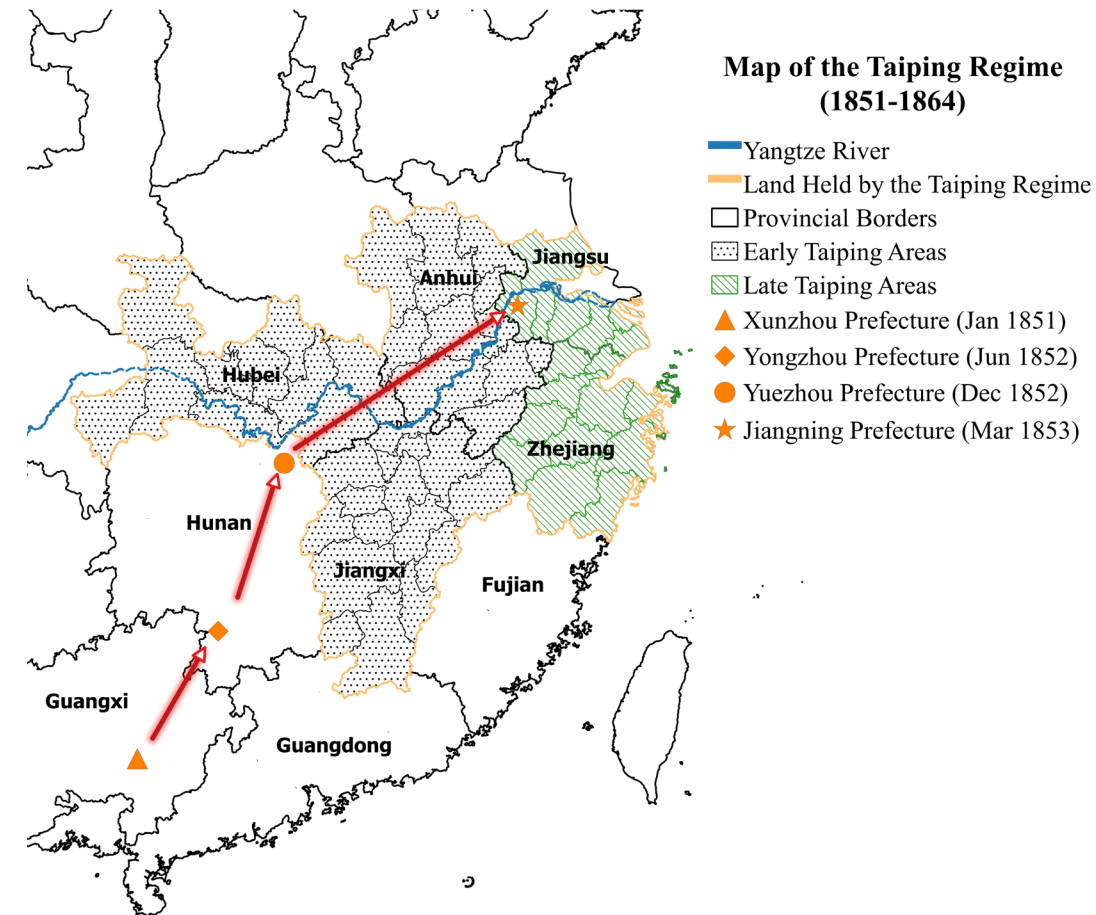
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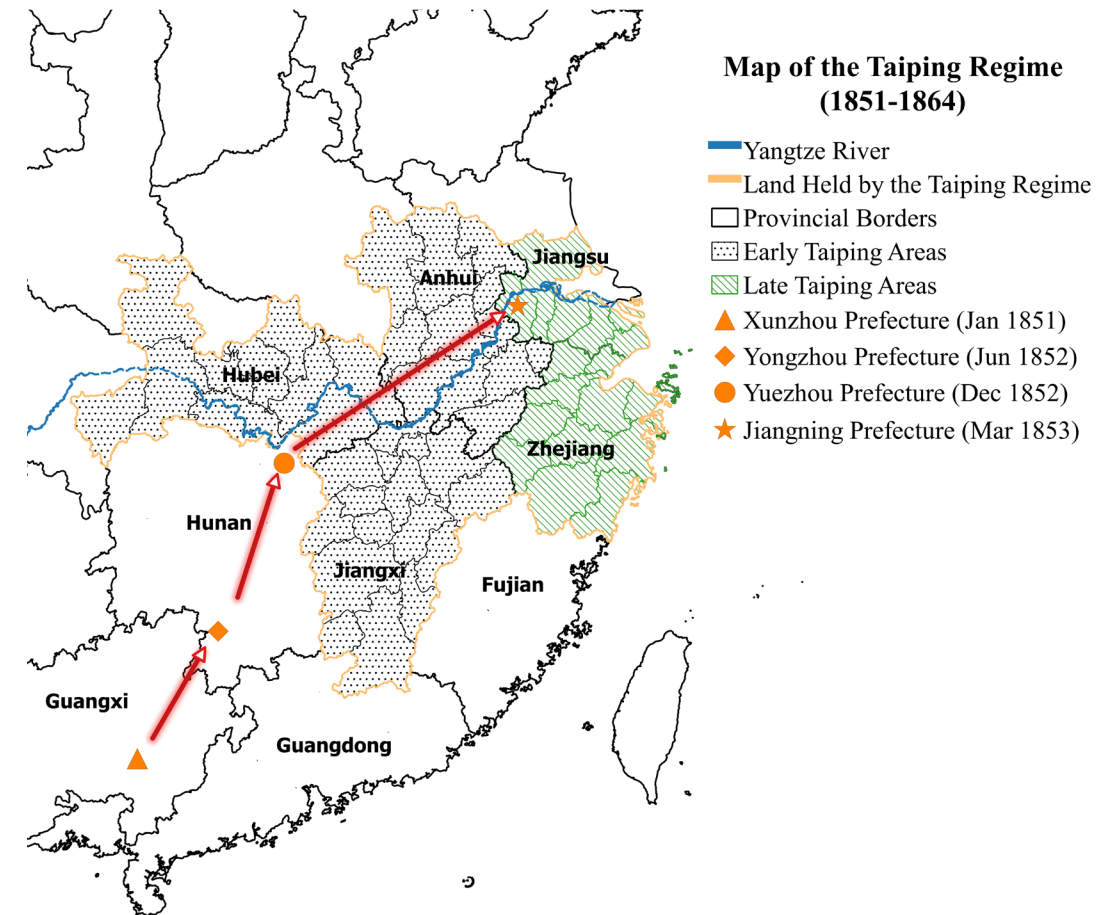
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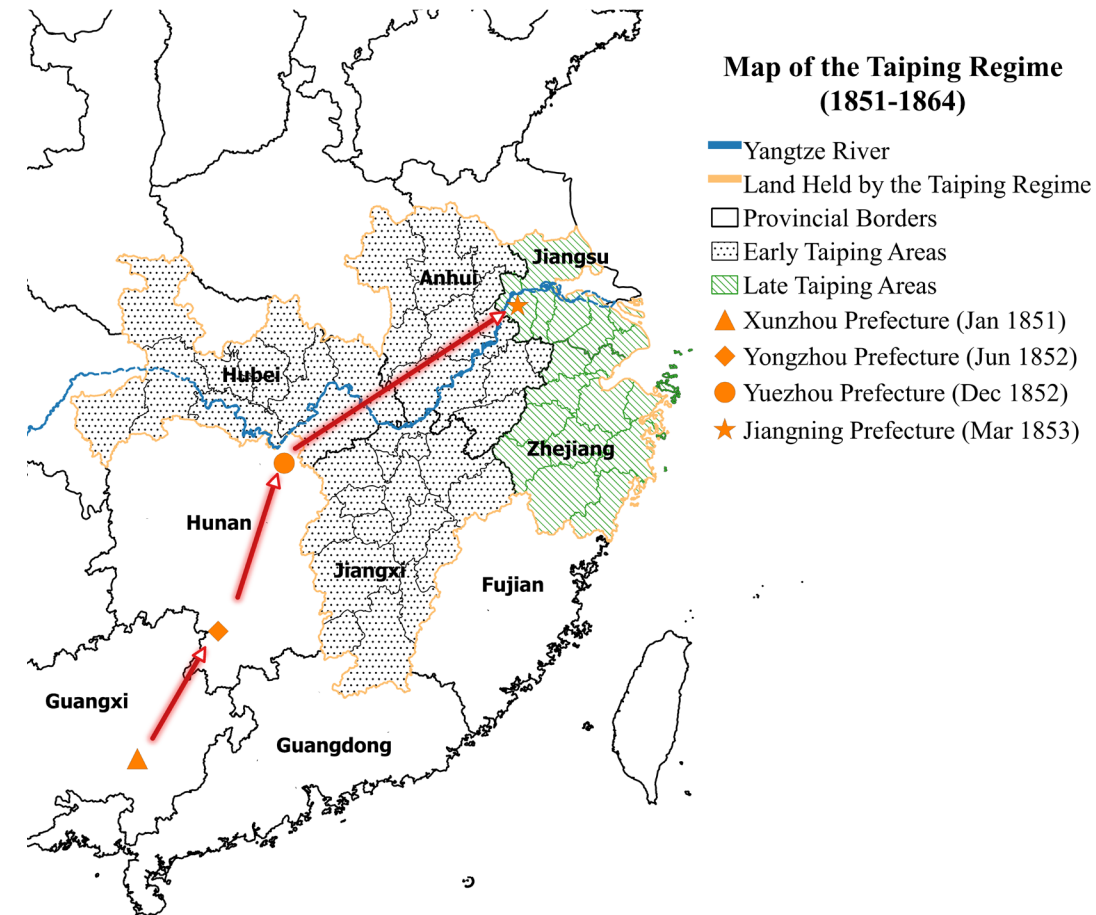
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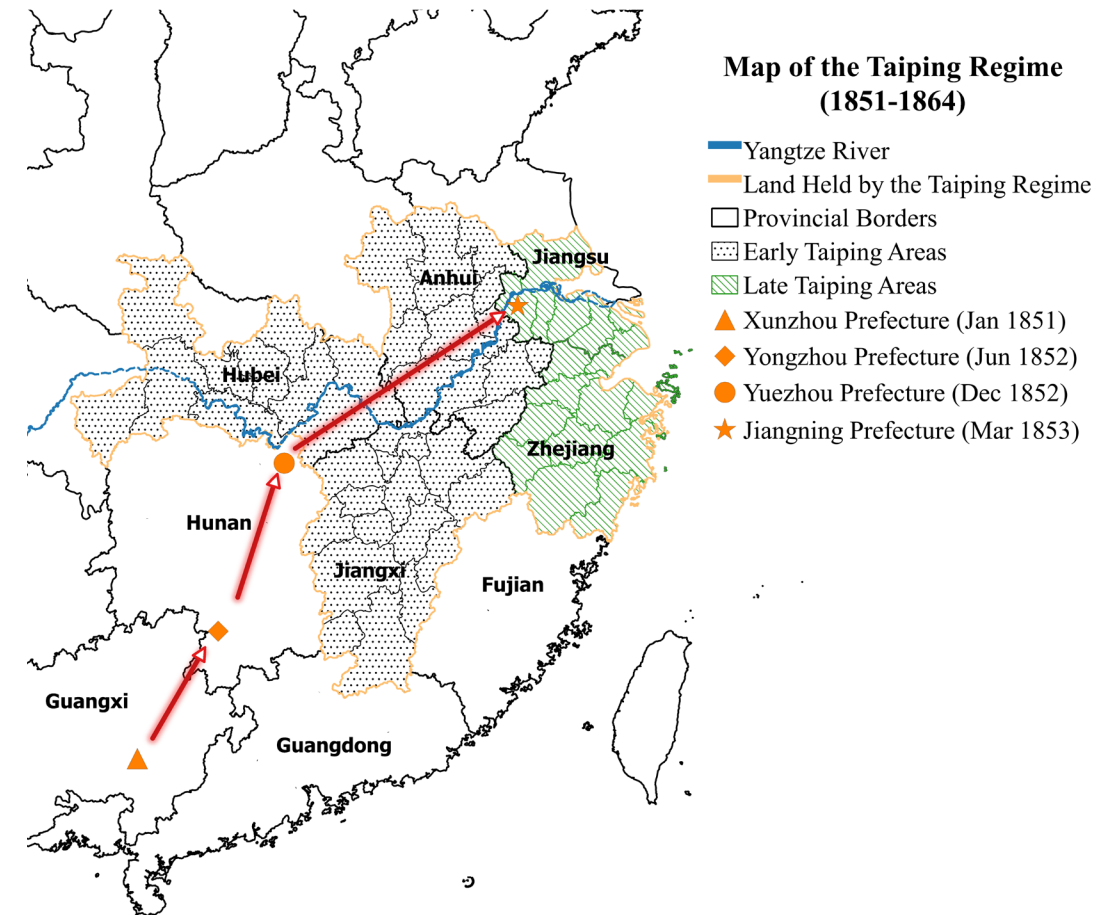
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  - **SOCIAL CAPACITY HYPOTHESIS:** The Rebellion would lead to stronger social cohesion (e.g., ↑interpersonal trust & political engagement)
    - (e.g., Acemoglu and Robinson. 2019; Woolcock and Narayan. 2000; Gittel and Vidal, 1998)



# Research Design

- We construct a panel of 266 Chinese prefectures from 1820-2000.
- We estimate the following panel regression model:

$$Y_{it} = \sum_{\tau \neq \text{reference yr. before rebellion}} \beta_{\tau} (Taiping_i \times T_{\tau}) + X_i' \gamma_t + \lambda_i + \mu_t + \varepsilon_{it}$$

Taiping dummy      Year dummy      Baseline covariates      Prefecture & Year FEs      Clustered by prefecture

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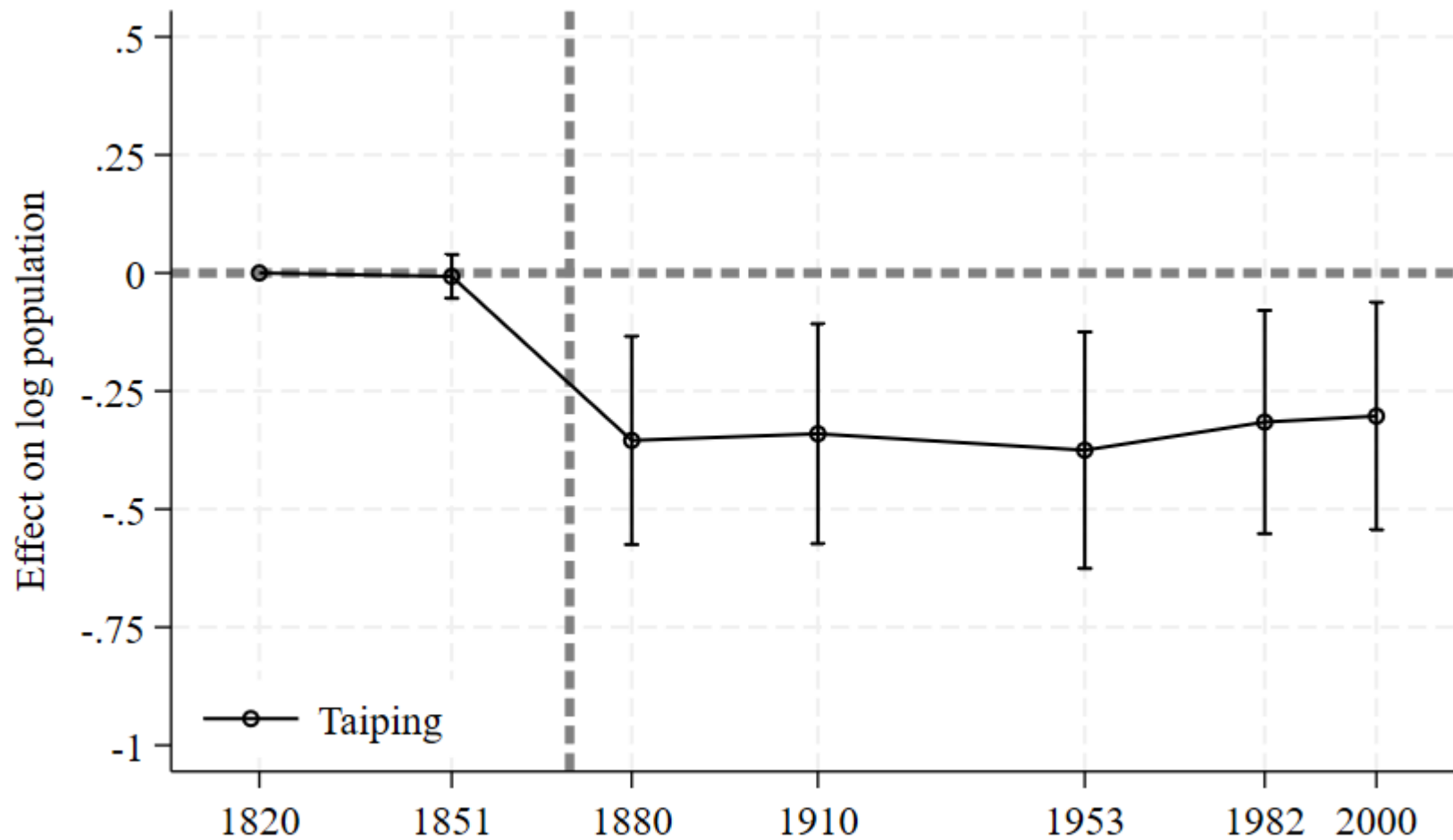
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- Sometimes also run a cross-section regression (with matching):

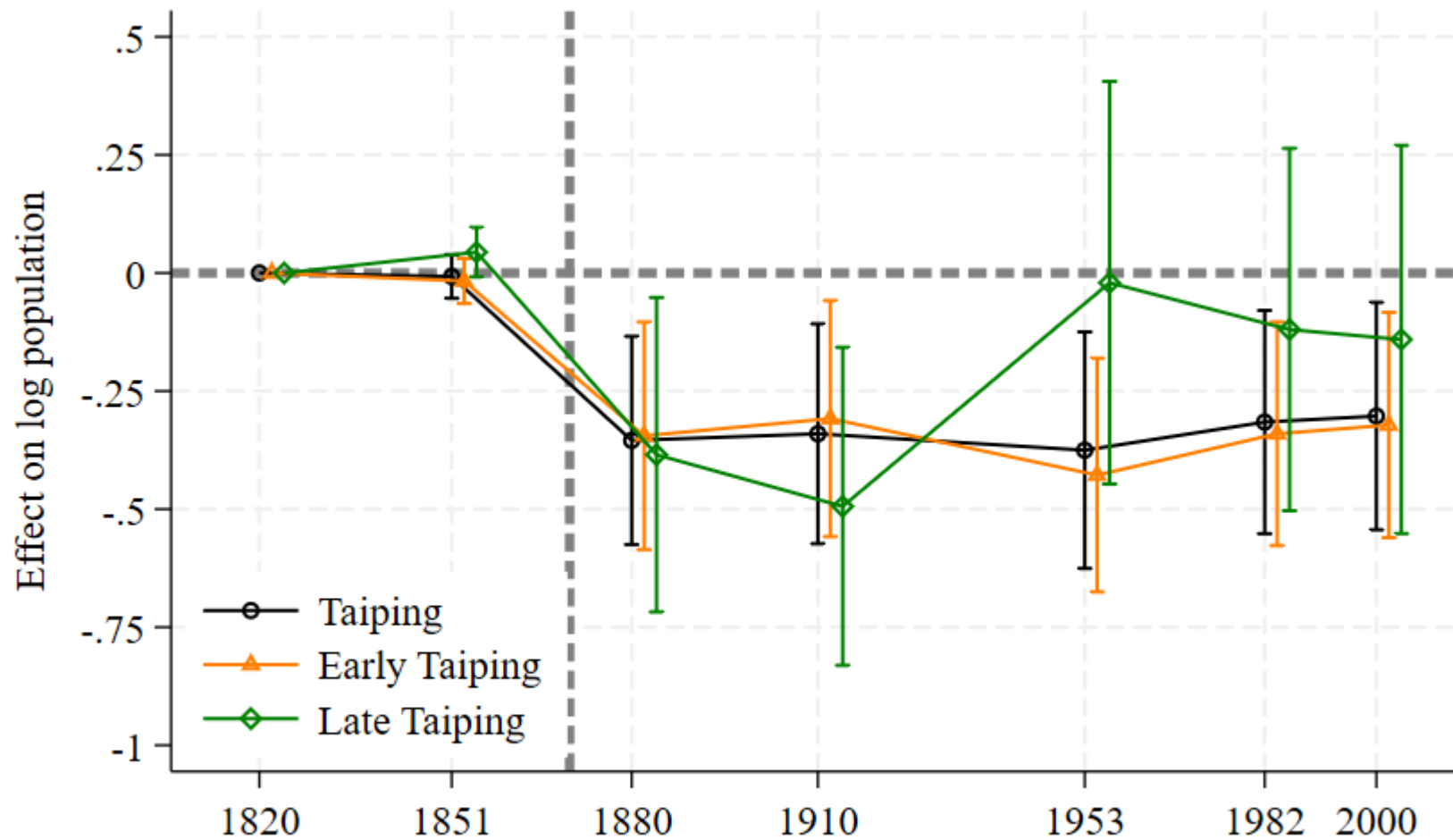
$$Y_i = \alpha + \beta Taiping_i + X_i' \gamma + \varepsilon_i$$

# Institutional Changes

# Effects on Population

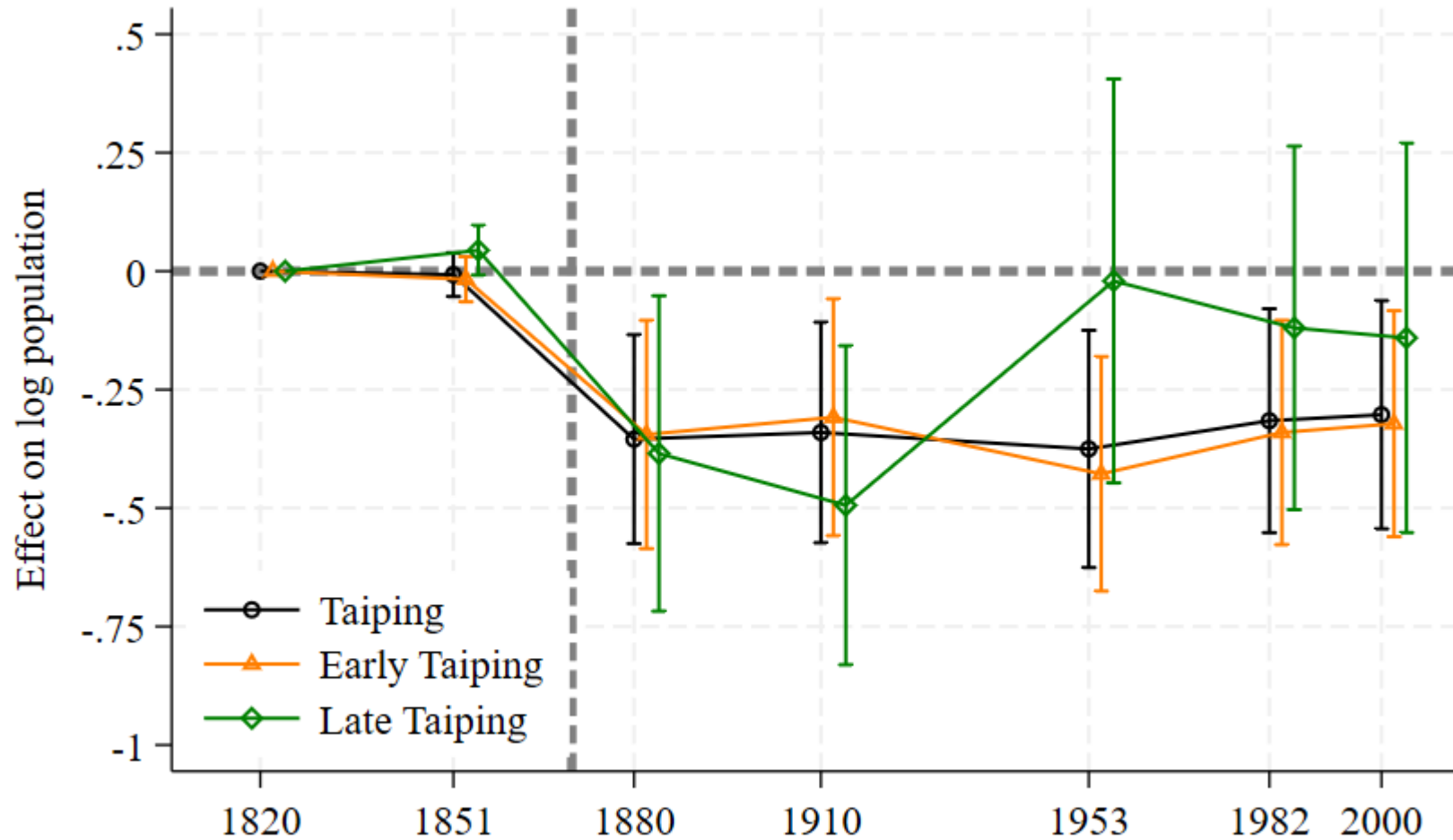


# Effects on Population



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On average, the Rebellion led to long-term population loss. However, **Late Taiping areas** rebounded.



# Testing Property Rights Hypothesis

**Panel B. Determinants of the share of idle land in 1915**

|                                      | (1)                | (2)                 | (3)                 |
|--------------------------------------|--------------------|---------------------|---------------------|
| Taiping                              | 2.710**<br>[1.034] |                     |                     |
| Early Taiping                        |                    | 3.555***<br>[1.278] | 3.855***<br>[1.327] |
| Late Taiping                         |                    | 1.26<br>[1.437]     | 1.692<br>[1.493]    |
| Standardized ln(distance to Nanjing) |                    |                     | -1.519<br>[1.252]   |
| Initial Likin                        |                    |                     | -1.025<br>[0.701]   |
| Observations                         | 332                | 332                 | 332                 |
| R-squared                            | 0.108              | 0.11                | 0.079               |
| Control                              | Y                  | Y                   | Y                   |
| Province fixed effect                | Y                  | Y                   | N                   |

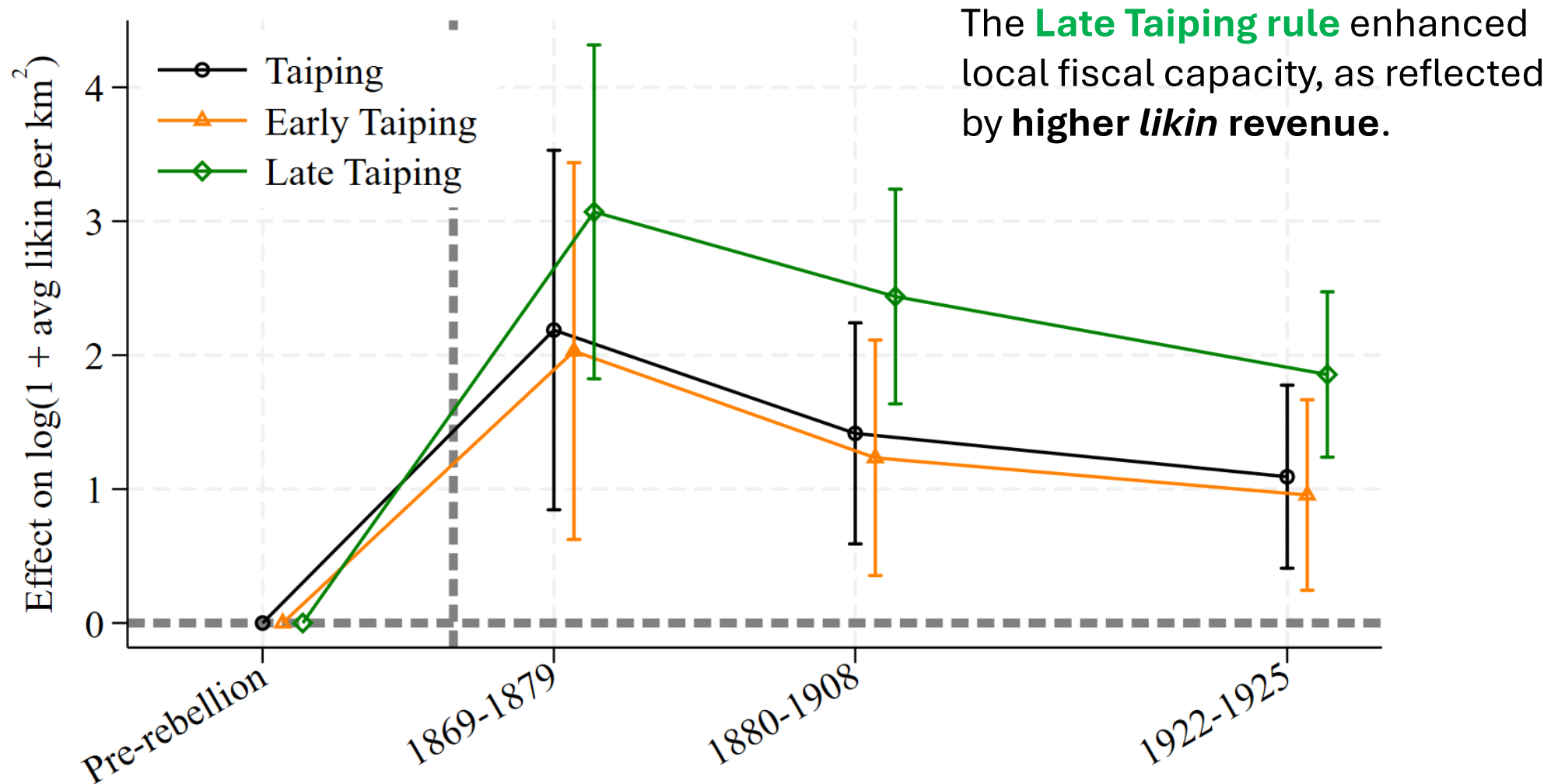
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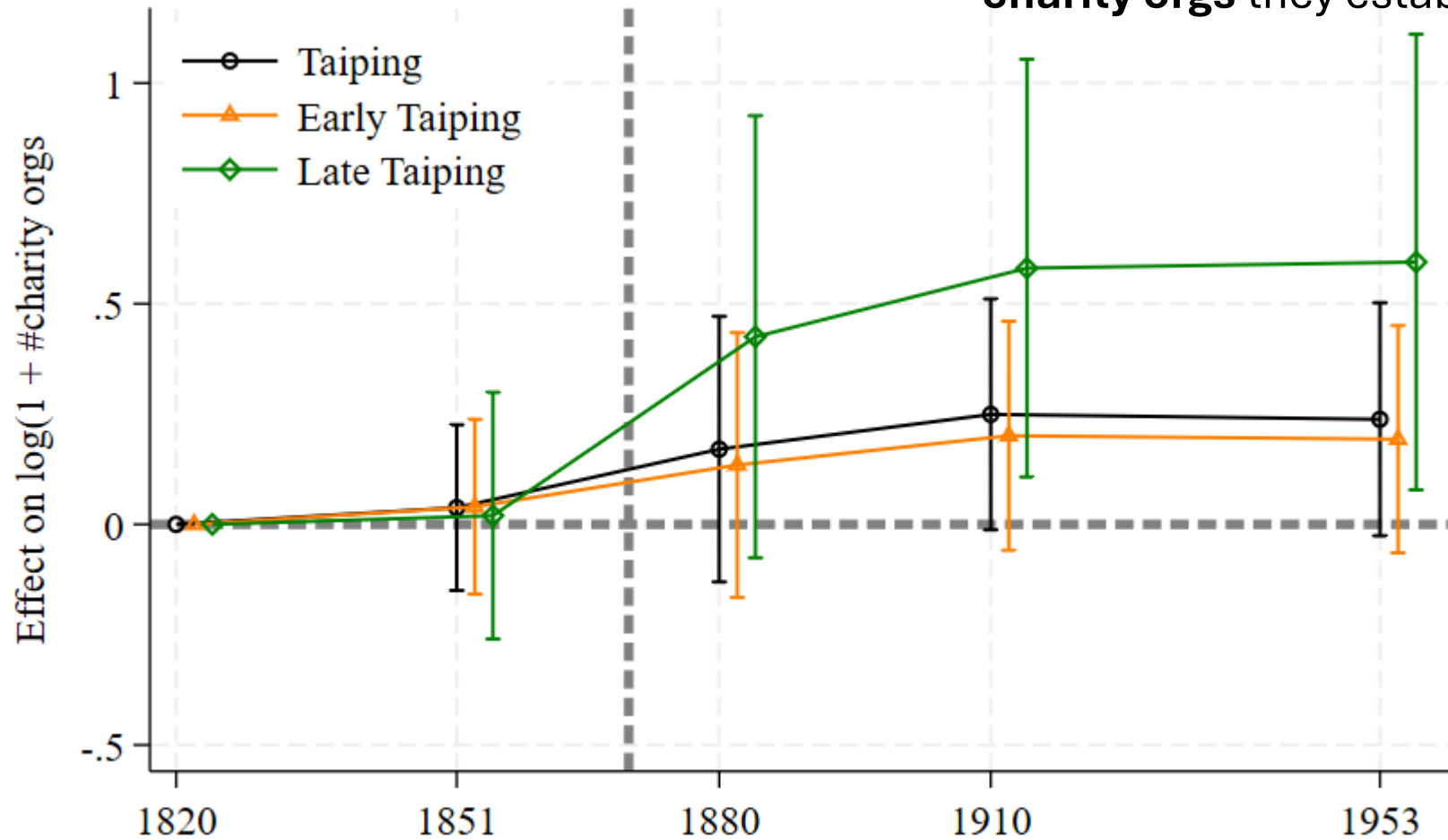
In **Late Taiping areas**, better property rights  
 → incentivize investment & maintenance  
 → no difference in land reclamation

# Testing Fiscal Capacity Hypothesis



# Testing Social Capacity Hypothesis

In **Late Taiping areas**, local elites are more engaged in local governance, as reflected by **more charity orgs** they established.



## **Long-Term Impacts**

# Long-Term Effects on Economic Development

| Dependent variable:               | Log ag & ind. output per capita (1982)<br>(1) | % Middle school graduates (1982)<br>(2) | % Literate population (1982)<br>(3) | Log avg. years of schooling (2000)<br>(6) | Log GDP per capita (2010)<br>(4)  | Log fiscal revenue per capita (2010)<br>(5) |
|-----------------------------------|-----------------------------------------------|-----------------------------------------|-------------------------------------|-------------------------------------------|-----------------------------------|---------------------------------------------|
| Early Taiping                     | 0.130<br>[0.091]                              | -0.007<br>[0.007]                       | -0.022<br>[0.020]                   | 0.019<br>[0.017]                          | -0.154<br>[0.105]                 | 0.190<br>[0.165]                            |
| <b>Late Taiping</b>               | <b>0.642***</b><br><b>[0.162]</b>             | <b>0.038***</b><br><b>[0.013]</b>       | <b>0.006</b><br><b>[0.029]</b>      | <b>0.025</b><br><b>[0.031]</b>            | <b>0.625***</b><br><b>[0.166]</b> | <b>1.109***</b><br><b>[0.222]</b>           |
| Ln (Jinshi <sub>1793-1820</sub> ) | 0.174***<br>[0.035]                           | 0.013***<br>[0.003]                     | 0.031***<br>[0.008]                 | 0.027***<br>[0.009]                       | 0.120**<br>[0.049]                | 0.201***<br>[0.053]                         |
| Ln (pre-TP wars)                  | 0.045<br>[0.113]                              | -0.002<br>[0.008]                       | -0.010<br>[0.020]                   | -0.014<br>[0.023]                         | -0.148<br>[0.098]                 | -0.263*<br>[0.137]                          |
| Controls                          | Y                                             | Y                                       | Y                                   | Y                                         | Y                                 | Y                                           |
| Observations                      | 191                                           | 191                                     | 191                                 | 191                                       | 185                               | 185                                         |
| R-squared                         | 0.437                                         | 0.339                                   | 0.279                               | 0.336                                     | 0.447                             | 0.479                                       |

- The **Late Taiping rule** increased income, fiscal revenue, and educational attainment.
- (In the paper) Controlling #charity orgs **attenuates** the Late Taiping effects → Part of the development effects are explained by stronger social capacity.

# Long-Term Effects on Sociopolitical Development

|                     | (1)                              | (2)                             | (3)                               | (4)                             | (5)                               |
|---------------------|----------------------------------|---------------------------------|-----------------------------------|---------------------------------|-----------------------------------|
| Dependent variable: | Trust in<br>Relatives            | Trust in<br>Friends             | Trust in<br>Coworkers             | Attention to<br>Politics        | Engagement<br>in Local<br>Affairs |
| Early Taiping       | 0.138***<br>[0.050]              | 0.072<br>[0.056]                | 0.077<br>[0.058]                  | -0.043*<br>[0.025]              | -0.021<br>[0.031]                 |
| <b>Late Taiping</b> | <b>0.142**</b><br><b>[0.069]</b> | <b>0.142*</b><br><b>[0.078]</b> | <b>0.188***</b><br><b>[0.066]</b> | <b>0.080*</b><br><b>[0.045]</b> | <b>0.079*</b><br><b>[0.042]</b>   |
| Observations        | 6416                             | 6404                            | 6123                              | 6406                            | 6415                              |
| R-squared           | 0.022                            | 0.025                           | 0.026                             | 0.297                           | 0.047                             |
| Baseline controls   | Y                                | Y                               | Y                                 | Y                               | Y                                 |
| Individual controls | Y                                | Y                               | Y                                 | Y                               | Y                                 |
| Cultural Revolution | Y                                | Y                               | Y                                 | Y                               | Y                                 |

The **Late Taiping rule** enhanced social cohesion and civic engagement.

# Conclusions

- Modern-day regional inequality can be rooted in the experience of conflict.
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